Total No. of Questions : 6]

SEAT No. :

P5789

[Total No. of Pages : 2

TE/INSEM/OCT. - 1
T.E. (Civil)
Hydrology and Water Resources Engineering
(2012 Pattern) (Semester - I)

Time : 1 Hour] *[Max. Marks :30*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) State the components of Hydrologic cycle. Explain three forms of precipitation. **[5]**

b) What is Dalton's law? Explain double ring infiltrometer. **[5]**

OR

Q2) a) Explain double mass curve of rainfall. **[5]**

b) A storm with 10 cm precipitation produced a direct runoff of 5.8 cm in a basin. Calculate average infiltration rate of the storm. **[5]**

Rain fall - Hrs	1	2	3	4	5	6	7	8
Incremental Rain fall (cm)	0.4	0.9	1.5	2.3	1.8	1.6	1.0	0.5

Q3) a) Define Duty. State the relationship between Duty and Delta. **[5]**

b) State various methods of irrigation. Write merits and demerits of Sprinkler Irrigation. **[5]**

P.T.O.

OR

Q4) a) Explain the following : **[5]**

i) Lacustrine soil

ii) OMC

iii) Kor Watering

iv) Delta

v) Cumecday

b) State various methods of canal revenue. Explain volumetric assessment of Canal Irrigation. **[5]**

Q5) a) Explain the following : **[5]**

i) Aquitard

ii) Specific yield

iii) Drawdown

iv) Transmissivity

v) Porosity

b) Write short notes on : **[5]**

i) Recuperation Test

ii) Cavity Type Tube Well.

OR

Q6) a) During recuperation test the water level in an open well was depressed by pumping by 3 m and it recuperated to 1.2 m in 2 hrs. Determine the yield from a well of 4 m diameter under a depression head of 2.5 m. **[5]**

b) Write down the assumptions in Dupuit - Theim theory. Explain interference among wells. **[5]**

♦♦♦♦